

Applied Optimization

Solution to Exercise 4 - Duality

Lagrange Duality (0.5 pt)

Consider the optimization problem

$$\begin{aligned} & \text{minimize} && x^2 + 1 \\ & \text{subject to} && (x - 2)(x - 4) \leq 0 \end{aligned}$$

with variable $x \in \mathbb{R}$.

(a) Analysis of primal problem: give the feasible set, the optimal value, and the optimal solution.

(b) Lagrangian and dual function: plot the objective $x^2 + 1$ versus x . On the same plot, show the feasible set, optimal point and value, and plot the Lagrangian $L(x, \lambda)$ versus x for a few positive values of λ . Verify the lower bound property ($p^* \geq \inf_x L(x, \lambda)$ for $\lambda \geq 0$). Compute and sketch the Lagrange dual function g .

(c) Lagrange dual problem: state the dual problem, and verify graphically that it is a concave maximization problem. Find the dual optimal value and dual optimal solution λ . Does strong duality hold?

Solution:

(a) The feasible set is the interval $[2, 4]$. The (unique) optimal point is $x^* = 2$, and the optimal value is $p^* = 5$.

(b) Figure 1 shows plots of f_0 and f_1 . The Lagrangian is $L(x, \lambda) = f_0 + \lambda f_1 = (1 + \lambda)x^2 - 6\lambda x + (1 + 8\lambda)$. Figure 2 shows the Lagrangian $L(x, \lambda)$ as a function of x for different values of $\lambda \geq 0$. Note that the minimum value of $L(x, \lambda)$ over x (i.e., $g(\lambda)$) is always less than or equal to p^* . It increases as λ varies from 0 toward 2, reaches its maximum at $\lambda = 2$, and then decreases again as λ increases above 2. We have equality $p^* = g(\lambda)$ for $\lambda = 2$. For $\lambda > -1$, the Lagrangian reaches its minimum at $\tilde{x} = 3\lambda/(1 + \lambda)$ where the gradient over x is 0. For $\lambda \leq -1$ it is unbounded below. Thus

$$g(\lambda) = \begin{cases} -9\lambda^2/(1 + \lambda) + 1 + 8\lambda & \lambda > -1 \\ -\infty & \lambda \leq -1 \end{cases} \quad (1)$$

which is plotted in 3. We can verify that the dual function is concave, that its value is equal to $p^* = 5$ for $\lambda = 2$, and less than p^* for other values of λ .

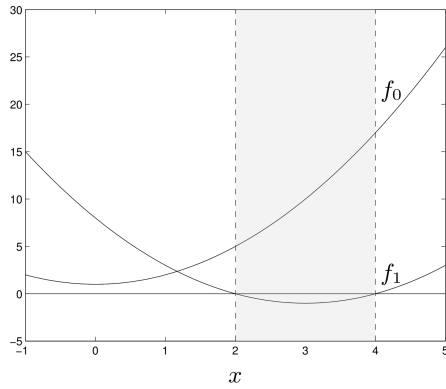


Figure 1: plot of objective and feasible set.

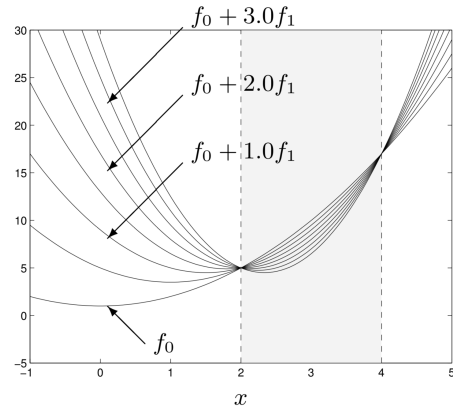


Figure 2: plot of Lagrangian

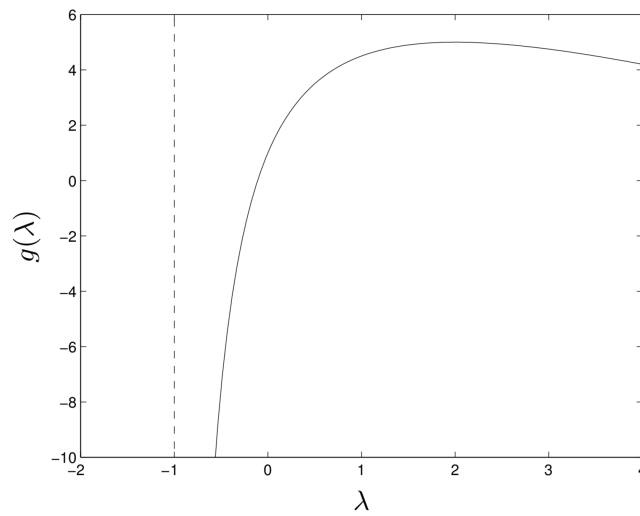


Figure 3: plot of $g(\lambda)$

(c) The Lagrange dual problem is

$$\begin{aligned} & \text{maximize} && -9\lambda^2/(1+\lambda) + 1 + 8\lambda \\ & \text{subject to} && \lambda \geq 0 \end{aligned} \tag{2}$$

The dual optimum occurs at $\lambda = 2$, with $d^* = 5$. So for this example we can directly observe that strong duality holds (as it must – Slater’s constraint qualification is satisfied).

KKT Condition (0.5 pt)

Consider the following optimization problem:

$$\begin{aligned} & \text{minimize} && x_1^2 - 2x_2^2 \\ & \text{subject to} && (x_1 + 4)^2 - 2 \leq x_2 \\ & && x_1 - x_2 + 4 = 0 \\ & && x_1 \geq -10 \end{aligned}$$

- (1) sketch the problem and graphically determine the primal solution x^* .
- (2) verify your x^* by determining suitable λ^* and ν^* such that the KKT conditions are satisfied for (x^*, λ^*, ν^*) .

Solution:

From the Figure 4, one could see that the optimum value is -4 at point $(-2, 2)$. Let us check the KKT conditions:

1. primal constraints:

$$(x_1 + 4)^2 - 2 - x_2 = (-2 + 4)^2 - 2 - 2 = 0$$

$$x_1 - x_2 + 4 = -2 - 2 + 4 = 0$$

$$-10 - x_1 = -10 - (-2) = -8 \leq 0$$

2. dual constraints: $\lambda_2 = 0 \geq 0$

3. complementary slackness: $\lambda_1(x_1 + 4)^2 - 2 - x_2 = 0, \lambda_2(-10 - x_1) = 0$

4. vanishing gradient of Lagrangian:

$$\begin{aligned} \nabla_x \mathcal{L}(x, \lambda, \nu) &= (2x_1, -4x_2)^T + \lambda_1(2x_1 + 8, -1)^T + \lambda_2(-1, 0)^T + \nu(1, -1)^T = (-4, -8)^T + \\ & 4(4, -1)^T - 12(1, -1)^T = (0, 0)^T \end{aligned}$$

We can then compute the dual variables $\lambda_1 = 4, \lambda_2 = 0$ and $\nu = -12$ from conditions (2), (3) and (4).

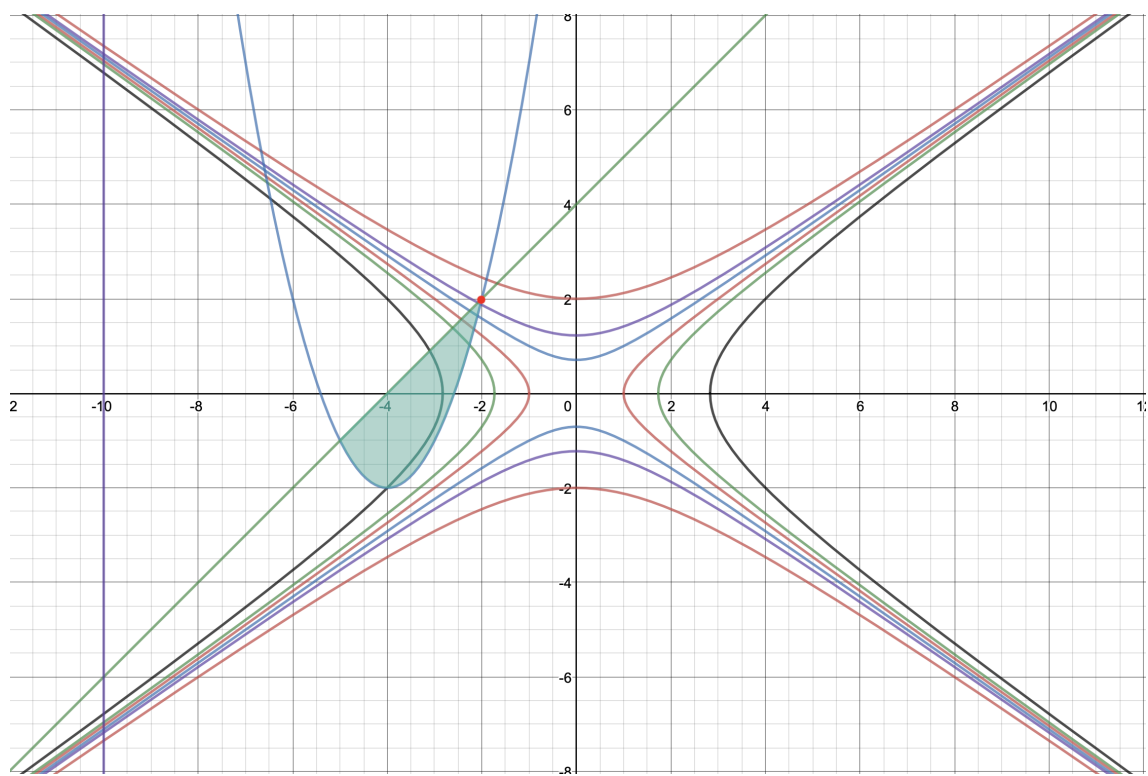


Figure 4: Sketch of the optimization problem